ANALYSIS II — SHEET 3 (for Week 4 tutorials)
The IVT. Boundedness Theorem.

1.(i) Show that if $f\colon \mathbb{C} \to \mathbb{R}$ is continuous such that $f(z_1) = -1$ and $f(z_2) = 1$ for $z_1, z_2 \in \mathbb{C}$, then $f(z) = 0$ for some $z \in \mathbb{C}$.

(ii) Provide an example of a continuous function $f\colon \mathbb{C} \to \mathbb{C}$ such that $f(0) = -1$ and $f(1) = 1$ and $f(z) = 0$ has no solution in $[0, 1]$. [Hint: consider quadratic f.]

(iii) Provide an example of a continuous function $f\colon \mathbb{C} \to \mathbb{C}$ such that $f(0) = -1$ and $f(1) = 1$ and $f(z) = 0$ has no solution at all in $\mathbb{C}$.

2. Let $p(x)$ be a polynomial of odd degree with positive leading coefficient.

(i) Show that $p(x) \to \pm\infty$ as $x \to \pm\infty$.

(ii) Carefully deduce that $p(x)$ has a real root.

3. How many continuous functions $f\colon \mathbb{R} \to \mathbb{R}$ are there such that

$$(f(x))^2 = x^2$$

for all real x? Carefully justify your answer. [Hint: consider the function $f(x)/x$ on $(0, \infty)$.]

4. (i) Let $f(x) = -3x^3 + x^2 + 12x - 4$ for $0 \leqslant x \leqslant 1$. Show that $f(0) < 0 < f(1)$.

(ii) Find explicit formulae for the sequences (a_n) and (b_n) constructed in the alternative 'bisection' proof of the IVT.

(iii) Determine their shared limit.

5. Give an example of a bounded, continuous function $f\colon [0, 1) \to \mathbb{R}$ which has neither a minimum nor a maximum.

6. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous function such that $f(x) \to \infty$ as $x \to \pm\infty$. Show that f is bounded below and achieves its minimum.

7. Let $n = 1, 2, 3\ldots$ and define $f_n\colon [0, \pi/2] \to \mathbb{R}$ by $f_n(x) = \max\{x^n, \cos x\}$.

(i) Show that f_n is bounded, achieves its bounds, and that f has a unique global minimum.

(ii) Say that minimum is achieved at α_n. Show that α_n satisfies the equation $(\alpha_n)^n = \cos\alpha_n$.

(iii) Show that (α_n) is increasing and that $\alpha_n \to 1$ as $n \to \infty$.

Extension Exercises (Optional)

8. Let $f\colon [0,1] \to \mathbb{R}$ be continuous with $f(0) = f(1)$. Prove that there exists $c \in [0, 1/2]$ such that

$$f(c) = f(c + 1/2).$$

9. Show that there is no continuous function $f\colon \mathbb{R} \to \mathbb{R}$ such that $|f^{-1}(y)| = 2$ for each $y \in f(\mathbb{R})$.

Construct a discontinuous function with this property.

10. Show that a continuous function $f\colon [a,b] \to \mathbb{R}$ is bounded as follows. Let

$$S = \{x \in [a,b] : f \text{ is bounded on } [a,x]\}.$$

(i) Show that $S \neq \varnothing$ and set $c = \sup S$.

(ii) Show that if $x \in S$ and $x < b$, then $x + \delta \in S$ for some $\delta > 0$.

(iii) Deduce that f is bounded on $[a,b]$.

11 Let $X \subseteq \mathbb{R}$ be such that every continuous function $f\colon X \to \mathbb{R}$ is bounded. Show that X is closed and bounded.

12. Given $f\colon [a,b] \to \mathbb{R}$ the *total variation* $V_{[a,b]}(f)$ is defined as

$$V_{[a,b]}(f) = \sup \left\{ \sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| \;\middle|\; a = x_0 < x_1 < \cdots < x_n = b \right\},$$

should the supremum be finite and otherwise $V_{[a,b]}(f) = \infty$. When $V_{[a,b]}(f) < \infty$ we say that f has *bounded variation.*

(i) Show that $f\colon [0,1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x \cos\left(\frac{\pi}{2x}\right) & 0 < x \leqslant 1; \\ 0 & x = 0, \end{cases}$$

is continuous, but does not have bounded variation.

(ii) Show that a monotonic function $f\colon [a,b] \to \mathbb{R}$ has bounded variation.

(iii) Show that if $f, g\colon [a,b] \to \mathbb{R}$ have bounded variation then so do $f + g$ and fg.